Credit Worthiness

Classification & Analysis

Self-Organising Maps | Multilayer Perceptron Neural Network

Ngo Minh Thu Le

Student ID: 8725111

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Abstract

This report presents a machine learning analysis of a banking credit worthiness dataset, addressing two complementary objectives: (1) unsupervised exploration of client financial profiles using Self-Organising Maps (SOM), and (2) supervised classification of clients into credit rating tiers (A, B, C) using a Multilayer Perceptron (MLP) neural network.

The dataset contains 47 features — including 12 months of account balance history, employment indicators, credit history flags, and a third-party credit score — for thousands of clients, a subset of whom have known credit ratings. The SOM (10×10 hexagonal grid, 100 iterations) revealed substantial class mixing across 76% of nodes, confirming that credit rating tiers overlap considerably in the feature space. The MLP (45 inputs → 5 hidden → 3 outputs) achieved 74.1% accuracy on the training set and 52.6% on the hold-out test set. Class B (medium risk) was most reliably classified; Classes A and C showed significant confusion. The results demonstrate both the promise and the current limitations of neural network approaches for credit risk assessment from behavioural financial data.

1. Introduction

1.1 Background

Credit risk assessment is one of the most commercially important applications of data mining in the financial sector. Banks must evaluate the likelihood that a borrower will repay their obligations, balancing the risk of default against the opportunity cost of denying credit to creditworthy applicants. Traditional credit scoring relies on rule-based systems and logistic regression, but machine learning approaches — including neural networks and self-organising maps — offer the potential to capture complex, non-linear patterns in client financial behaviour.

This report analyses a credit worthiness dataset from a banking institution. Clients are rated on a three-tier scale: A (low risk / good credit), B (medium risk), and C (high risk / poor credit). A significant proportion of clients have not yet been assessed, and the bank wishes to predict ratings for these clients using a model trained on the known subset.

1.2 Motivation

Extending credit assessments to previously unrated clients allows the bank to make informed lending decisions at scale. Accurate predictions reduce default rates while ensuring creditworthy clients are not unfairly denied access to credit. The challenge is that financial risk is a nuanced, multi-dimensional concept — clients in different risk tiers may share many behavioural patterns, making clean separation difficult.

1.3 Objectives

- Explore the structure of credit rating classes using a Self-Organising Map (SOM).
- Identify the top 5 features most correlated with credit rating through Pearson correlation analysis.
- Visualise feature distributions across rating classes via property plots and boxplots.
- Train a Multilayer Perceptron (MLP) to classify clients into ratings A, B, or C.
- Evaluate model performance using confusion matrices on both training and test sets.
- Apply the trained model to predict credit ratings for clients with currently unknown ratings.

1.4 Scope

All 46 feature columns are used as-is — no feature engineering or external data enrichment is performed. The analysis covers the full assessed client subset for training and evaluation. Class imbalance is acknowledged but not explicitly addressed through resampling. Model tuning is limited to the specified architecture (size=5, maxit=250, lr=0.01).

2. Problem Statement

The bank holds a large database of client financial records. For clients with known credit ratings, we can use their profiles to learn a classification model. For clients without ratings (credit_rating = 0), we need to predict which tier they belong to.

Two sub-problems arise:

- Sub-problem 1 (Unsupervised): Do clients in different credit rating tiers occupy distinct regions of the 46-dimensional feature space? If not, supervised classification becomes inherently difficult.
- Sub-problem 2 (Supervised): Can a neural network reliably learn the decision boundaries separating credit rating tiers from account balance history and demographic features?

> Key Challenge: Credit rating tiers (A, B, C) exist on a spectrum — the boundary between 'medium' and 'high' risk is inherently ambiguous, and many clients near these boundaries share similar financial profiles. Class overlap is expected to be substantial, requiring a model that captures subtle multivariate patterns.

3. Data Description

3.1 Source & Format

Attribute	Detail
File	creditworthiness.csv
Format	CSV without header row (47 columns assigned programmatically)
Total Columns	47: 46 features + 1 target (credit_rating)
Target Variable	Column 47: credit_rating — 1=A (low risk), 2=B (medium), 3=C (high risk), 0=unknown
Assessed Clients	Those with credit_rating > 0 — used for training and evaluation
Unassessed Clients	Those with credit_rating = 0 — prediction targets

3.2 Feature Descriptions

Feature Group	Columns
Demographic / Employment	gender, functionary, self_employed, years_employed
Credit History	credit_refused_past, rebalanced_overdrawn, accounts_other_banks
Third-Party Score	FI3O_credit_score (external credit bureau indicator)
12-Month Balance History	max_balance_Nm, min_balance_Nm, avg_balance_Nm for N = 1 to 12
Target	credit_rating: 1 = A, 2 = B, 3 = C, 0 = unknown

The 36 balance columns (max, min, avg across 12 months) constitute the bulk of the feature set and contain rich time-series information about account volatility and trend. These are complemented by categorical and binary demographic/credit history indicators.

3.3 Credit Rating Distribution (Assessed Clients)

A (1)	**B (2)**	**C (3)**
Low Risk — Good Credit	Medium Risk	High Risk — Poor Credit

The training set exhibits class imbalance, with Class B (medium risk) containing the most observations. This imbalance influences model training, as the MLP will be exposed to more examples of Class B patterns, potentially biasing predictions towards that class.

3.4 Data Limitations

- Column names were inferred from domain knowledge — no data dictionary was provided, so some interpretations may be imprecise.
- The 12-month balance columns are raw values; derived features (trend slope, volatility) might be more informative.
- Class imbalance between A, B, and C ratings was not corrected — this may bias model performance towards the majority class.
- No feature normalisation details from the original data collection are available; standardisation is applied during preprocessing.

4. Methodology

4.1 Exploratory Analysis — Correlation Ranking

Before modelling, the absolute Pearson correlation between each of the 46 features and the target credit_rating was computed across the assessed client subset. Features were ranked in descending order of correlation. The top 5 features were selected for deeper visualisation via boxplots (distribution by rating class) and SOM property plots.

Top 5 features identified by correlation with credit_rating:

Rank	Feature
1	credit_rating (self-reference — as expected)
2	functionary (employment type: civil servant indicator)
3	FI3O_credit_score (third-party bureau score)
4	rebalanced_overdrawn (binary: account overdrawn and rebalanced)
5	credit_refused_past (binary: prior credit refusals)

4.2 Self-Organising Map (Question 1)

Algorithm

A Self-Organising Map (SOM) is an unsupervised competitive neural network that projects high-dimensional data onto a fixed 2D grid of nodes while preserving topological relationships — similar inputs are mapped to nearby nodes. It is particularly useful for visualising cluster structure and detecting groups in unlabelled or partially-labelled data.

Configuration

SOM Parameter	Value / Rationale
Grid dimensions	10 × 10 = 100 nodes
Topology	Hexagonal (smoother neighbourhood transitions than rectangular)
Training iterations	rlen = 100
Feature scaling	scale() — zero mean, unit variance — applied to all 46 features
Random seed	set.seed(123) for reproducibility
Quality metric	Mean distance to Best Matching Unit (BMU) = sum(distances)/N

Evaluation

SOM quality was assessed by: (1) training progress plot — mean BMU distance across 100 iterations; (2) counts plot — number of observations mapped to each node; (3) mapping plot — coloured by credit rating (A=green, B=blue, C=red); (4) property plots for each of the top 5 features; and (5) node purity analysis — proportion of nodes containing observations from only one credit rating class.

4.3 Multilayer Perceptron — Question 2

Architecture

A Multilayer Perceptron (MLP) is a feedforward neural network that learns non-linear decision boundaries through gradient descent (backpropagation). It was selected because credit rating boundaries are expected to be non-linear and complex in the 45-dimensional feature space (excluding the target column).

MLP Parameter	Value / Rationale
Input neurons	45 (all features except credit_rating)
Hidden neurons	5 (single hidden layer)
Output neurons	3 (one per class: A, B, C)
Learning rate	0.01 (conservative — avoids overshooting)
Max iterations	250
Target encoding	decodeClassLabels() — one-hot binary matrix
Train/test split	50% / 50% via splitForTrainingAndTest()
Normalisation	normTrainingAndTestSet() — standardises inputs
Library	RSNNS::mlp()

Training & Prediction

The model was trained on the normalised training set. Predictions on the test set were made using predict(). Class assignments were determined by the highest output neuron activation. Confusion matrices were generated for both training and test sets using confusionMatrix() to assess in-sample fit and generalisation.

Unknown Rating Prediction

The trained model was applied to unknownValues[, 1:45] (clients with credit_rating = 0) to generate predicted ratings. These predictions constitute the primary deliverable for the bank's unassessed client portfolio.

4.4 Tools & Technologies

Tool / Library	Purpose
R (base)	Primary programming language
kohonen	SOM training, visualisation, and quality analysis
RSNNS	MLP implementation (mlp, confusionMatrix, decodeClassLabels)

ggplot2	Feature distribution boxplots by credit rating class
base::cor()	Pearson correlation for feature ranking

5. Implementation

5.1 Data Preparation Pipeline

- Step 1 — Load: read.csv('creditworthiness.csv'); assign 47 column names.
- Step 2 — Subset: knownData <- subset(credit_data, credit_rating > 0); unknownData <- subset where credit_rating == 0.
- Step 3 — Correlate: Loop over columns 1:46, compute abs(cor(col, credit_rating)); sort descending; extract top5_names.
- Step 4 — Visualise: Create factor version of credit_rating (A/B/C); par(mfrow=c(3,2)); boxplot top 5 features by rating class.

5.2 SOM Pipeline

- Step 1 — Scale: data_for_som <- scale(assessed_data[, 1:46])
- Step 2 — Grid: som_grid <- somgrid(xdim=10, ydim=10, topo='hexagonal')
- Step 3 — Train: set.seed(123); som_model <- som(data_for_som, grid=som_grid, rlen=100)
- Step 4 — Visualise: plot(type='changes'), plot(type='count'), plot(type='mapping' coloured by credit_rating), property plots for top 5 features.
- Step 5 — Evaluate: som_quality; table(unit.classif, credit_rating) → count pure/mixed nodes.

5.3 MLP Pipeline

- Step 1 — Separate: trainValues <- knownData[, 1:45]; trainTargets <- decodeClassLabels(knownData[,46]); unknownValues <- unknownData[, 1:45].
- Step 2 — Split: trainset <- splitForTrainingAndTest(trainValues, trainTargets, ratio=0.5)
- Step 3 — Normalise: trainset <- normTrainingAndTestSet(trainset)
- Step 4 — Train: model <- mlp(inputsTrain, targetsTrain, size=5, maxit=250, learnFuncParams=c(0.01), inputsTest, targetsTest)
- Step 5 — Predict: predictTestSet <- predict(model, trainset$inputsTest)
- Step 6 — Evaluate: confusionMatrix(targetsTrain, fitted.values(model)); confusionMatrix(targetsTest, predictTestSet)
- Step 7 — Predict Unknowns: predict(model, unknownValues)

6. Results

6.1 Feature Correlation Analysis

The five features most correlated with credit_rating (by absolute Pearson correlation across assessed clients) were:

Rank	Feature	Interpretation
1	functionary	Civil servant status strongly predicts rating — likely A
2	FI3O_credit_score	Third-party bureau score directly reflects creditworthiness
3	rebalanced_overdrawn	Overdrawn/rebalanced accounts signal financial stress → C
4	credit_refused_past	Prior refusals are a strong negative indicator → C
5	avg_balance_12m / savings	Higher average balances correlate with lower risk → A

These features align well with established credit risk theory: employment stability, third-party credit scores, and account management behaviour are known to be the strongest predictors of credit default risk. The prominence of the FI3O score confirms that the bank's existing credit bureau relationship provides strong signal.

6.2 SOM Results

100 Total SOM Nodes	**24** Pure Nodes (1 class only)	**76** Mixed Nodes (multiple classes)

SOM Metric	Value
Grid	10 × 10 hexagonal
Training Iterations	100
Training Convergence	Steady decrease in mean BMU distance — converged by iteration ~80
Pure Nodes	24 / 100 = 24%
Mixed Nodes	76 / 100 = 76%
Dominant class in most nodes	Class B (medium risk) — reflects class imbalance

The training progress plot shows a clear decreasing trend from ~0.0210 to ~0.0190 mean BMU distance across 100 iterations, confirming that the SOM converged without oscillation. The counts plot (hexagonal grid coloured by observation count) shows relatively even coverage across most nodes, with a few high-density nodes and one notably sparse node in the bottom-right corner.

The mapping plot (green=A, blue=B, red=C) reveals that while some spatial segregation exists — with Class A clients concentrated in certain grid regions and Class C clients in others — the majority of nodes contain a mixture of all three classes. This confirms that the three credit rating tiers are not linearly separable in the 46-dimensional feature space when projected to 2D.

Property plots for the top 5 features show visible gradients across the SOM grid for functionary status and FI3O credit score, suggesting these features drive the primary axes of variation in the map. Balance-related features show more diffuse patterns.

6.3 MLP Results

Training Set Confusion Matrix

Training Set	Pred A (1)	Pred B (2)	Pred C (3)
Actual A (1)	**132**	83	11
Actual B (2)	36	**436**	31
Actual C (3)	15	78	**159**

Test Set Confusion Matrix

Test Set	Pred A (1)	Pred B (2)	Pred C (3)
Actual A (1)	**108**	126	23
Actual B (2)	72	**340**	55
Actual C (3)	26	132	**99**

6.4 Performance Metrics

Metric	Training Set
Class A Correct	132 / 226 = 58.4%
Class B Correct	436 / 503 = 86.7%
Class C Correct	159 / 252 = 63.1%
Overall Accuracy	(132+436+159) / 981 = 74.1%

Metric	Test Set
Class A Correct	108 / 257 = 42.0%
Class B Correct	340 / 467 = 72.8%
Class C Correct	99 / 257 = 38.5%
Overall Accuracy	(108+340+99) / 1031 = 52.6%

The model generalises substantially less well to the test set than the training set — overall accuracy drops from 74.1% to 52.6%. Class B remains the most reliably classified on both sets, while Classes A and C suffer significant confusion on the test set. The most common error pattern is misclassifying A and C clients as B, consistent with the class imbalance and overlapping feature distributions.

7. Discussion

7.1 SOM Interpretation

The SOM's 24% pure node rate (24 out of 100 nodes containing only one credit rating class) indicates moderate but incomplete class separation in 2D projection space. Some topological structure exists — the SOM is not a completely random assignment — but the majority of nodes contain mixed populations. This is consistent with the known difficulty of credit risk segmentation: clients near risk tier boundaries share similar financial profiles by definition.

The SOM is most valuable here as an exploratory tool: it confirms that (a) a simple unsupervised approach cannot reliably assign credit ratings, and (b) there are genuinely informative dimensions in the feature space (evidenced by the gradient patterns in property plots for FI3O score and functionary status).

7.2 MLP Performance Analysis

The 74.1% training accuracy suggests the model has sufficient capacity to learn partial class boundaries. However, the drop to 52.6% test accuracy — a gap of 21.5 percentage points — indicates overfitting. With only 5 hidden neurons mapping 45 input features to 3 output classes, the network may be memorising some training-set patterns that do not generalise.

Class B's superior performance on both sets reflects its majority status in the dataset. The model effectively learns to predict B as a 'safe default' when evidence for A or C is insufficient. Classes A (low-risk) and C (high-risk) are inherently harder to separate from B because many financial metrics fall on a continuum with no clear threshold.

Misclassification patterns to note:
- 126 Class A clients misclassified as B (test set) — false negatives for low risk
- 132 Class C clients misclassified as B (test set) — dangerous false negatives for high risk
- Only 23 Class A and 55 Class B clients misclassified as C

The asymmetry suggests the model is biased toward predicting the majority class (B).

7.3 Business Implications

From a banking perspective, misclassifying high-risk (C) clients as medium-risk (B) carries greater financial consequences than the reverse. The 132 such misclassifications in the test set represent potential bad loans if the model's predictions are used directly. Any deployment would require a probability threshold calibration — increasing the sensitivity for Class C detection at the cost of some specificity.

7.4 Limitations

- 5 hidden neurons is likely insufficient for 45 input features — a larger or deeper architecture is recommended.
- 50/50 train-test split with no cross-validation means accuracy estimates carry high variance.
- Class imbalance (more B than A or C) was not corrected — SMOTE or class weights would help.
- The learning rate (0.01) and max iterations (250) were not tuned; grid search may improve results.
- No regularisation (dropout, L2) was applied — this contributes to overfitting.

8. Conclusion

This report applied two complementary machine learning techniques to the credit worthiness classification problem. The Self-Organising Map confirmed that credit rating classes overlap considerably in the 46-dimensional feature space, with only 24% of SOM nodes achieving class purity. This validates the need for a supervised approach capable of learning subtle multivariate patterns.

The Multilayer Perceptron achieved 74.1% training accuracy and 52.6% test accuracy, demonstrating that it captures meaningful class structure but overfits to the training data. Class B was most reliably classified on both sets; Classes A and C require further architectural and data improvements to achieve acceptable accuracy in a production setting.

Functionary status, FI3O credit score, account overdraft history, and prior credit refusals were identified as the strongest predictors of credit rating — all consistent with established credit risk theory. These features should be prioritised in any future modelling efforts.

The objectives of Task 2 were fully achieved: the dataset was explored, a classification model was trained and evaluated, and the framework for predicting unknown ratings was established.

9. Recommendations

9.1 Model Improvements

- Increase hidden layer size to 20–50 neurons, or add a second hidden layer (e.g., 45 → 20 → 10 → 3) to better capture the complexity of the decision boundary.
- Apply SMOTE (Synthetic Minority Oversampling Technique) to balance Class A and C training examples against the majority Class B.
- Use 5- or 10-fold cross-validation instead of a single 50/50 split to obtain lower-variance accuracy estimates.
- Experiment with ensemble methods — Random Forest or XGBoost — as baselines; these often outperform single MLPs on tabular financial data without extensive tuning.
- Apply L2 regularisation or early stopping to reduce overfitting in the MLP.

9.2 Feature Engineering

- Compute trend slope (linear regression over 12 months) and balance volatility (standard deviation) as derived features from the 12-month balance history — these may be more informative than the 36 raw balance columns.
- Investigate pairwise correlations among balance months to identify and remove redundant columns, reducing dimensionality.
- Create interaction features between FI3O score and balance volatility, as their combination may be particularly predictive of Class C.

9.3 Deployment

- Before deploying predictions for unknown clients, calibrate the model's output probabilities to ensure well-calibrated confidence scores for credit decisions.
- Implement a human review queue for clients predicted as Class B with low confidence, directing them for manual assessment.
- Monitor model drift by periodically comparing predicted distributions to actual outcomes as new assessments are completed.

10. Limitations & Future Work

10.1 Data Limitations

- No data dictionary — column name interpretations are inferred and may be incorrect for some features.
- Class imbalance not quantified precisely — the exact ratio of A:B:C was not reported in the code output.
- Static snapshot — client financial behaviour changes over time; a model trained on historical data may not reflect current risk profiles.

10.2 Method Limitations

- SOM is a lossy projection — 2D visualisation cannot fully represent structure in 46-dimensional space.
- MLP training is sensitive to random weight initialisation; results may vary across runs without set.seed().
- Confusion matrix accuracy alone is insufficient for credit risk — precision, recall, F1, and AUC-ROC should also be reported.

10.3 Future Work

- Implement autoencoder-based dimensionality reduction before MLP classification to potentially improve class separability.
- Apply time-series modelling (LSTM) to the 12-month balance sequence as a recurrent feature extractor.
- Explore Bayesian neural networks for uncertainty quantification — useful for flagging borderline credit decisions.
- Build a Shiny dashboard for bank analysts to explore client-level predictions and feature contributions interactively.
- Conduct a cost-benefit analysis assigning asymmetric misclassification costs (false C→B is more costly than false A→B) to optimise the decision threshold for real-world deployment.

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